

Raphael E. Hviding

Postdoctoral Researcher | Max-Planck Institute for Astronomy | He/His

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Research Interests: Active Galactic Nuclei, Data Science, (Slitless) Spectroscopy

APPOINTMENTS

Max-Planck Institute for Astronomy

Heidelberg, DE

Postdoctoral Researcher

Sep. 2023 - pres.

- *RUBIES Team under Dr. Anna de Graaff*
 - *Statistical analyses of Little Red Dots with JWST NIRSpec.*
- *Data Science Group under Dr. Ivelina Momcheva*
 - *Slitless Spectroscopic Reduction & Analysis for (Parallel) Extragalactic Surveys*
- *Collaborations: RUBIES, OutThere, 3D-DASH, and Euclid*

EDUCATION

University of Arizona

Tucson, AZ

Ph.D., M.Sc. in Astronomy & Astrophysics

Aug. 2023, Jan. 2021

- *Seeing **Red**: The Present and Future of mid-IR AGN Selection with Optical Spectroscopy*
- *Advisors: Professor Kevin N. Hainline & Professor Marcia J. Rieke*

Dartmouth College

Hanover, NH

B.A. in Physics (High Honors), Mathematics, and Astronomy (Minor)

Jun. 2018

- *Senior Honors Thesis under Professor Ryan C. Hickox*

WORKSHOPS

AI Agents Tutorial

Jan. 2026

- Tutorial on GAI in scientific workflows and integrations with IDEs and MPC servers.

Pixi Tutorial

Jan. 2026

- Tutorial on cross-platform reproducible (Python) environment management.

HPC Workshop

Mar. 2024, 2025

- Workshop covering parallelization (threading and multiprocessing), SLURM scripting, containers (Docker/Apptainer), and workflow managers (Snakemake).

TEACHING

ASTR 170B: Exploring our Universe

UArizona

Teaching Assistant with Professor Ed Prather

Spring 2023

- Participated in lesson activities and led office hours for a general-education course.

ASTR 400B: Theoretical Astrophysics: Galaxies and Cosmology

UArizona

Teaching Assistant with Professor Dan Stark

Spring 2021

- Led office hours, graded, and taught a lecture for an advanced undergraduate course.

MENTORING

MPIA Summer Internship

Heidelberg, DE

Summer Intership Co-Mentor

Jun. 2025 - Sep. 2025

– Co-advised a summer intern in their investigation of dwarf galaxies in NIRISS imaging.

Summer Intership Mentor

Jun. 2024 - Sep. 2024

– Advised a summer intern in their investigation of obscured AGN candidate spectra.

NOAO Astronomy Teen Café

Tucson, AZ

Graduate Student Guest and Discussion Leader

Oct. 2018 - May. 2022

– Worked with high school students on astronomical coding exercises and college advice.

Project ASTRO

Tucson, AZ

Astronomer Collaborator of a School Teacher

Sep. 2018 - May. 2018

– Worked with a primary school teacher to plan astronomical lessons for the classroom.

PRESENTATIONS

Invited Talks.....

Nov. 2025 **Tel Aviv University**, Astro Seminar Tel-Aviv, IL

May. 2024 **MPIA**, Königstuhl Colloquium Heidelberg, DE

Jun. 2024 **Durham University**, Special Talk Durham, UK

Workshops.....

Feb. 2026 **ISSI Team 25-659** Bern, CH

Jan. 2026 **CSI: Sesto** Sexten, IT

Conferences.....

Apr. 2026 **PyCon DE & PyData 2026**, Talk Darmstadt, DE

May. 2025 **Crisol 2025**, Talk Toledo, ES

Apr. 2025 **PyCon DE & PyData 2025**, Talk Darmstadt, DE

Jan. 2025 **Euclid GAEV Meeting**, Talk Tenerife, ES

Jan. 2023 **AAS 241**, Talk Seattle, WA

Oct. 2022 **SACNAS NDiSTEM 2022**, Talk San Juan, PR

Nov. 2018 **Advances with SALT**, Talk Pretoria, SA

Sep. 2022 **What drives the growth of black holes?**, Poster Reykjavík, IS

Jun. 2019 **IAU Galaxies 2019**, Poster Viana do Castelo, PT

Nov. 2017 **Ivy League Undergraduate Research Symposium**, Poster Philadelphia, PA

Aug. 2016 **Hidden Monsters**, Poster Hanover, NH

Jun. 2016 **Active Galactic Nuclei: What's in a name?**, Poster Garching, DE

PRESS

Selected Media Coverage.....

Feb. 2026 **Scientific American**, *Weird new object escalates ‘black hole star’ debate*

Jan. 2026 **Astrobites**, *A Potential New Piece of the “Little Red Dot” Puzzle*

Sep. 2025 **Max-Planck-Gesellschaft**, *Are Black Hole Stars real?*

Feb. 2020 **University of Arizona**, *On Student Success, This Astronomer Walks the Walk*

Dec. 2017 **Dartmouth College**, *Studying the Stars in the South African Sky*

CODE DEVELOPMENT

unite

Lead Developer

Unified Numerical Integration Tool for spEctroscopy

Python

- *Joint Spectroscopic Fitting with NumPyro and JAX.*
- *Used to jointly fit multiple (NIRSpec) dispersers while accounting for undersampling.*

GELATO

Lead Developer

Galaxy/AGN Emission Line Analysis TOol

Python

- *Optical Spectroscopic Fitting package with a focus on testing for AGN contributions.*
- *In use by the DESI Collaboration, a LEGA-C Collaboration project, Dr. Mar Mezcua's group at Institute of Space Sciences in Spain, and Dr. Kohei Ichikawa's group at Tohoku University.*

Grizli

Contributor

Grism redshift & line analysis software for space-based slitless spectroscopy

Python

- *Contributions specifically to improving performance and accuracy with NIRISS WFSS data.*

Programming: Python, Linux, IRAF, SQL, ADQL, HTML/CSS, git, L^AT_EX, SLURM

TELESCOPE TIME

Principal Investigator

NOEMA W25 *Probing Exotic Dust and Gas Obscuration in a High-Redshift, X-Ray Luminous AGN* (10h; Grade A)

LBTO 2024B *Seeing Double: A Survey of Dual and Lensed AGN at Cosmic Noon* (12h)

LBTO 2024A *Seeing Double: A Survey of Dual and Lensed AGN at Cosmic Noon (Pilot)* (5h)

SAO 2023A *Revealing Obscured SMBH Growth: Probing Type II AGN Candidates with MMT Binospec* with MMT Binospec (2 Nights)

SAO 2022B *Uncovering the True Nature of the Kiloparsec-Scale Ionization in NGC 1068: Past AGN Activity or Shocks?* with MMT MMIRS and Bok BCSpec (1+3 Nights)

SAO 2021A *MMIRS longslit follow-up of mid-IR AGN candidates with high Balmer Decrements: Obscured optical line emission?* with MMT MMIRS (1.5 Nights)

SAO 2020B *Uncovering an Undiscovered Population of AGNs: Hectospec Follow-up of HSC-WISE-SDSS Matched Targets* with MMT Hectospec (2 Nights)

Co-Investigator

MPG 2.2m 117 *Cracking the Egg: Precision Variability of a Little Red Dot* (92h)

HST Cycle 34 *Beyond the Quasar Redshift Frontier: Uncovering Rapidly Accreting Supermassive Black Holes at $z > 8$ with HST/WFC3 and Euclid* (SNAP – 125 Orbits)

NOEMA W24 *Extremely compact galaxies at Cosmic Dawn: ultra-massive galaxies or AGN?* (16h; Grade A)

HONORS & AWARDS

Awarded		
2022	Departmental Graduate Student Award for Service	UArizona
2018 - 2023	National Science Foundation Graduate Research Fellowship	NSF
2018	College of Science Fellowship	UArizona
2018	International Travel Grant	AAS
2017 - 2018	E. E. Just Scholar	Dartmouth College

PUBLICATIONS

Twelve major contributing author publications: [full list]

(*eight as first/corresponding author)

- [1] Anna de Graaff, **Raphael E. Hviding** et al., 2026, *MNRAS*, 551, [stag1567](#), *Little red dots host black hole stars: a unified family of gas-reddened AGN revealed by JWST/NIRSpec spectroscopy*
- [2] Jenny E. Greene, Hanpu Liu, **Raphael E. Hviding** et al., 2026, *arXiv*, [arXiv:2609.25349](#), *Calcium Triplet Absorption is Common around Little Red Dots*
- [*3] **Raphael E. Hviding** et al., 2026, *ApJL*, 1000, L18, *The X-Ray Dot: Exotic Dust or a Late-stage Little Red Dot?*
- [*4] **Raphael E. Hviding** et al., 2025, *A&A*, 702, A57, *RUBIES: A spectroscopic census of little red dots: All point sources with v-shaped continua have broad lines*
- [*5] **Raphael E. Hviding** et al., 2024, *AJ*, 168, 220, *Improved Empirical Backgrounds for JWST NIRISS Image/Wide-field Slitless Spectroscopy Data Reduction*
- [*6] **Raphael E. Hviding** et al., 2024, *AJ*, 167, 169, *Spectroscopic Confirmation of Obscured AGN Populations from Unsupervised Machine Learning*
- [*7] **Raphael E. Hviding** et al., 2023, *AJ*, 166, 111, *The Kiloparsec-scale Influence of the AGN in NGC 1068 with SALT RSS Fabry-Pérot Spectroscopy*
- [*8] **Raphael E. Hviding** et al., 2022, *AJ*, 163, 224, *A New Infrared Criterion for Selecting Active Galactic Nuclei to Lower Luminosities*
- [9] Kevin N. Hainline, **Raphael E. Hviding** et al., 2020, *ApJ*, 892, 125, *Simulating JWST/NIRCam Color Selection of High-redshift Galaxies*
- [10] L. Claire Gasque, Callum A. Hening, **Raphael E. Hviding** et al., 2019, *AJ*, 158, 156, *Two Long-period Cataclysmic Variable Stars: ASASSN-14ho and V1062 Cyg*
- [*11] **Raphael E. Hviding** et al., 2018, *ApJ*, 868, 16, *Spatially Extended Low-ionization Emission Regions (LIERs) at $z \sim 0.9$*
- [*12] **Raphael E. Hviding** et al., 2018, *MNRAS*, 474, 1955, *Characterizing the WISE-selected heavily obscured quasar population with optical spectroscopy from the Southern African Large Telescope*

46 total publications, 34 as contributing author:

- [1] Bingjie Wang, ..., **Raphael E. Hviding** et al., 2026, *arXiv*, [arXiv:2609.22623](#), *Balmer Absorption Series and Broad Metal Lines in Two Luminous Little Red Dots*
- [2] Rohan P. Naidu, ..., **Raphael E. Hviding** et al., 2026, *Nature*, 656, 329, *A gas-enshrouded and gas-reddened black hole at cosmic dawn*

- [3] Ravi Jaiswar, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2608.19687](#), *OutThere Survey: Addressing f_{ion} and f_{esc} with a population of average galaxies at $z \sim 2$*
- [4] C. Bertemes, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2608.10070](#), *A cosmic collision in the making: JWST/NIRISS reveals a merging and maturing protocluster core at $z \sim 3$ around a red quasar*
- [5] Andrea Weibel, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2606.17271](#), *Black Hole Stars Across the Universe: Identifying Central Engine Dominated Little Red Dots at $z \sim 1.5 - 9.5$*
- [6] Rohan P. Naidu, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2606.30711](#), *Little Red Dots as Intermediate Mass, Super-Eddington Engines: Insights from Type II In Supernovae and The 1837-1856 Great Eruption of Carinae*
- [7] Quinn O. Casey, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2606.26098](#), *A Population of Little Red Dot-like Quasars in SDSS*
- [8] Bingjie Wang, ..., **Raphael E. Hviding** et al., 2026, [ApJ](#), 1003, 10, *The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra Resemble Massive Stars*
- [9] Wendy Q. Sun, ..., **Raphael E. Hviding** et al., 2026, [OJAp](#), 9, 62505, *Little Red Dot – Host Galaxy = Black Hole Star: A Gas-Enshrouded Heart at the Center of Every Little Red Dot*
- [10] L. N. Martínez-Ramírez, ..., **Raphael E. Hviding** et al., 2026, [A&A](#), 708, A117, *16 new quasars at the end of the reionization unveiled by self-supervised learning*
- [11] Jorrryt Matthee, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2603.17667](#), *The Engine and its Flows: Little Red Dot spectra are shaped by the column densities of their gas envelopes*
- [12] Daniel J. Eisenstein, ..., **Raphael E. Hviding** et al., 2026, [ApJS](#), 283, 6, *Overview of the JWST Advanced Deep Extragalactic Survey (JADES)*
- [13] Aidan P. Cloonan, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2603.24700](#), *A PANORAMIC of UV-optical morphologies of “Little Red Dots”: Two groups of LRDs distinguished by UV half-light radius*
- [14] Bingjie Wang, ..., **Raphael E. Hviding** et al., 2026, [arXiv](#), [arXiv:2602.06024](#), *Water absorption confirms cool atmospheres in two little red dots*
- [15] Rohan P. Naidu, ..., **Raphael E. Hviding** et al., 2026, [OJAp](#), 9, 56033, *A Cosmic Miracle: A Remarkably Luminous Galaxy at $z_{\text{spec}} = 14.44$ Confirmed with JWST*
- [16] Jenny E. Greene, ..., **Raphael E. Hviding** et al., 2026, [ApJ](#), 996, 129, *What You See Is What You Get: Empirically Measured Bolometric Luminosities of Little Red Dots*
- [17] Rodrigo Córdova Rosado, ..., **Raphael E. Hviding** et al., 2025, [ApJ](#), 995, 227, *Cross-correlation of Luminous Red Galaxies with ML-selected Active Galactic Nuclei in HSC-SSP. II. AGN Classification and Clustering with DESI Spectroscopy*
- [18] Anna de Graaff, ..., **Raphael E. Hviding** et al., 2025, [A&A](#), 701, A168, *A remarkable ruby: Absorption in dense gas, rather than evolved stars, drives the extreme Balmer break of a little red dot at $z = 3.5$*
- [19] Anna de Graaff, ..., **Raphael E. Hviding** et al., 2025, [A&A](#), 697, A189, *RUBIES: A complete census of the bright and red distant Universe with JWST/NIRSpec*
- [20] Andrea Weibel, ..., **Raphael E. Hviding** et al., 2025, [ApJ](#), 983, 11, *RUBIES Reveals a Massive Quiescent Galaxy at $z = 7.3$*

- [21] Olivia R. Cooper, ..., **Raphael E. Hviding** et al., 2025, *ApJ*, 982, 125, *RUBIES: JWST/NIRSpec Resolves Evolutionary Phases of Dusty Star-forming Galaxies at $z \sim 2$*
- [22] Ragadeepika Pucha, ..., **Raphael E. Hviding** et al., 2025, *ApJ*, 982, 10, *Tripling the Census of Dwarf AGN Candidates Using DESI Early Data*
- [23] Charity Woodrum, ..., **Raphael E. Hviding** et al., 2024, *PNAS*, 121, e2317375121, *Using JADES NIRCam photometry to investigate the dependence of stellar mass inferences on the IMF in the early universe*
- [24] Charity Woodrum, ..., **Raphael E. Hviding** et al., 2024, *ApJ*, 974, 305, *Active Galactic Nuclei in the Green Valley at $z \sim 0.7$*
- [25] Andrew J. Bunker, ..., **Raphael E. Hviding** et al., 2024, *A&A*, 690, A288, *JADES NIRSpec initial data release for the Hubble Ultra Deep Field: Redshifts and line fluxes of distant galaxies from the deepest JWST Cycle 1 NIRSpec multi-object spectroscopy*
- [26] Morgan Fouesneau, ..., **Raphael E. Hviding** et al., 2024, *arXiv*, arXiv:2409.20252, *What is the Role of Large Language Models in the Evolution of Astronomy Research?*
- [27] Kevin N. Hainline, ..., **Raphael E. Hviding** et al., 2024, *ApJ*, 964, 66, *Brown Dwarf Candidates in the JADES and CEERS Extragalactic Surveys*
- [28] Kevin N. Hainline, ..., **Raphael E. Hviding** et al., 2024, *ApJ*, 964, 71, *The Cosmos in Its Infancy: JADES Galaxy Candidates at $z > 8$ in GOODS-S and GOODS-N*
- [29] Jakob M. Helton, ..., **Raphael E. Hviding** et al., 2024, *ApJ*, 962, 124, *The JWST Advanced Deep Extragalactic Survey: Discovery of an Extreme Galaxy Overdensity at $z = 5.4$ with JWST/NIRCam in GOODS-S*
- [30] Marcia J. Rieke, ..., **Raphael E. Hviding** et al., 2023, *ApJS*, 269, 16, *JADES Initial Data Release for the Hubble Ultra Deep Field: Revealing the Faint Infrared Sky with Deep JWST NIRCam Imaging*
- [31] B. E. Robertson, ..., **Raphael E. Hviding** et al., 2023, *NatAs*, 7, 611, *Identification and properties of intense star-forming galaxies at redshifts $z > 10$*
- [32] Emma Curtis-Lake, ..., **Raphael E. Hviding** et al., 2023, *NatAs*, 7, 622, *Spectroscopic confirmation of four metal-poor galaxies at $z = 10.3$ -13.2*
- [33] Marcia Rieke, ..., **Raphael E. Hviding** et al., 2019, *BAAS*, 51, 45, *JWST GTO/ERS Deep Surveys*
- [34] Wei Yan, ..., **Raphael E. Hviding** et al., 2019, *ApJ*, 870, 33, *NuSTAR and Keck Observations of Heavily Obscured Quasars Selected by WISE*